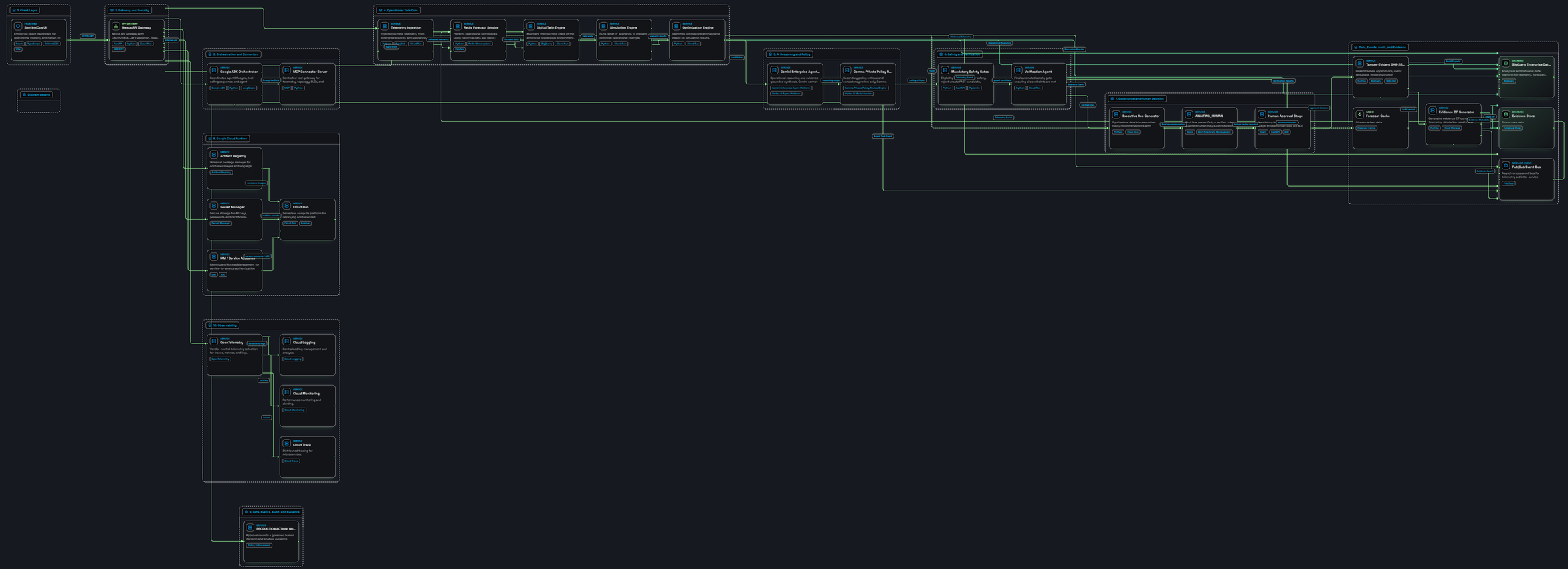

1. Client Layer
FRONTEND
SentinelOps UI
Enterprise React dashboard for operational visibility and human-in-
React
TypeScript
Tailwind CSS
Vite
HTTPS/JWT
2. Gateway and Security
API GATEWAY
Nexus API Gateway
Nexus API Gateway with OAuth2/OIDC, JWT validation, RBAC,
FastAPI
Python
Cloud Run
IAM/ADC
internal-api
Diagram Legend
3. Orchestration and Connectors
SERVICE
Google ADK Orchestrator
Coordinates agent lifecycle, tool-calling sequence, and agent
Google ADK
Python
LangGraph
Enterprise Data
SERVICE
MCP Connector Server
Controlled tool gateway for telemetry, topology, SLOs, and
MCP
Python
4. Operational Twin Core
SERVICE
Telemetry Ingestion
Ingests real-time telemetry from enterprise sources with validation
Python
Pub/Sub
Cloud Run
Sync State
validated telemetry
SERVICE
Redis Forecast Service
Predicts operational bottlenecks using historical data and Redis-
Python
Redis Memorystore
Pandas
forecast data
SERVICE
Digital Twin Engine
Maintains the real-time state of the enterprise operational environment.
Python
BigQuery
Cloud Run
twin state
SERVICE
Simulation Engine
Runs 'what-if' scenarios to evaluate potential operational changes.
Python
Cloud Run
scenario results
SERVICE
Optimization Engine
Identifies optimal operational paths based on simulation results.
Python
Cloud Run
candidates
5. AI Reasoning and Policy
SERVICE
Gemini Enterprise Agent...
Operational reasoning and evidence-grounded synthesis. Gemini cannot
Gemini Enterprise Agent Platform
Vertex AI Agent Platform
reasoning output
SERVICE
Gemma Private Policy R...
Secondary policy critique and consistency review only. Gemma
Gemma Private Policy Review Engine
Vertex AI Model Garden
policy critique
Agent Task Event
State
Historical Telemetry
Operational Analytics
Simulation Results
6. Safety and Verification
SERVICE
Mandatory Safety Gates
Eligibility Telemetry Event r safety, reject unsafe FAST candidate,
Python
FastAPI
Pydantic
gated candidate
SERVICE
Verification Agent
Final automated safety gate ensuring all constraints are met
Python
Cloud Run
Scenario Event
verified plan
Telemetry Event
Verification Results
7. Governance and Human Decision
SERVICE
Executive Rec Generator
Synthesizes data into executive-ready recommendations with
Python
Cloud Run
final recommendation
SERVICE
AWAITING_HUMAN
Workflow pause. Only a verified, role-qualified human may submit Accept
Redis
Workflow State Management
human review required
SERVICE
Human Approval Stage
Mandatory h Verification Event stage. Production actions are NOT
React
FastAPI
IAM
approval decision
Data, Events, Audit, and Evidence
SERVICE
Tamper-Evident SHA-25...
Linked hashes, append-only event sequence, model invocation
Python
BigQuery
SHA-256
Audit Exports
DATABASE
BigQuery Enterprise Dat...
Analytical and historical data platform for telemetry, forecasts,
BigQuery
CACHE
Forecast Cache
Stores cached data
Forecast Cache
audit record
SERVICE
Evidence ZIP Generator
Generates evidence ZIP cont telemetry, simulation results, and
Python
Cloud Storage
Store ZIP
Evidence Metadata
DATABASE
Evidence Store
Stores core data
Evidence Store
Evidence Event
MESSAGE-QUEUE
Pub/Sub Event Bus
Asynchronous event bus for telemetry and inter-service
Pub/Sub
8. Google Cloud Runtime
SERVICE
Artifact Registry
Universal package manager for container images and language
Artifact Registry
container images
SERVICE
Secret Manager
Secure storage for API keys, passwords, and certificates.
Secret Manager
runtime secrets
SERVICE
Cloud Run
Serverless compute platform for deploying containerized
Cloud Run
Knative
SERVICE
IAM / Service Accounts
service accounts / ADC
Identity and Access Management for service-to-service authentication
IAM
ADC
10. Observability
SERVICE
OpenTelemetry
structured logs
Vendor-neutral telemetry collection for traces, metrics, and logs.
OpenTelemetry
metrics
traces
SERVICE
Cloud Logging
Centralized log management and analysis.
Cloud Logging
SERVICE
Cloud Monitoring
Performance monitoring and alerting.
Cloud Monitoring
SERVICE
Cloud Trace
Distributed tracing for microservices.
Cloud Trace
9. Data, Events, Audit, and Evidence
SERVICE
PRODUCTION ACTION: NO...
Approval records a governed human decision and enables evidence
Policy Enforcement